NIH DATA SCIENCE & AI TAXONOMY

Impact of AI on Biomedical Research

Where AI shows up in NIH-funded grants, and what those grants produce: publications, clinical trials, and patents

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Research analytics at the intersection of biomedical research and data science

THE QUESTION

Funders can’t yet see where AI is used across NIH grants, or what it delivers

How much?

AI-related awards and funding over time

Where?

By institute, method, and research area

With what impact?

Publications, citations, trials, and patents

APPROACH

Four steps turn grant abstracts into validated AI labels

1 · Collect

Grant abstracts

NIH RePORTER, FY2015 to FY2026, with RCDC categories

→

2 · Define

AI taxonomy

Methods (ML, deep learning, NLP, LLMs) and applications

→

3 · Label

Embeddings + LLM

Cluster abstracts, then assign taxonomy labels with an LLM

→

4 · Validate

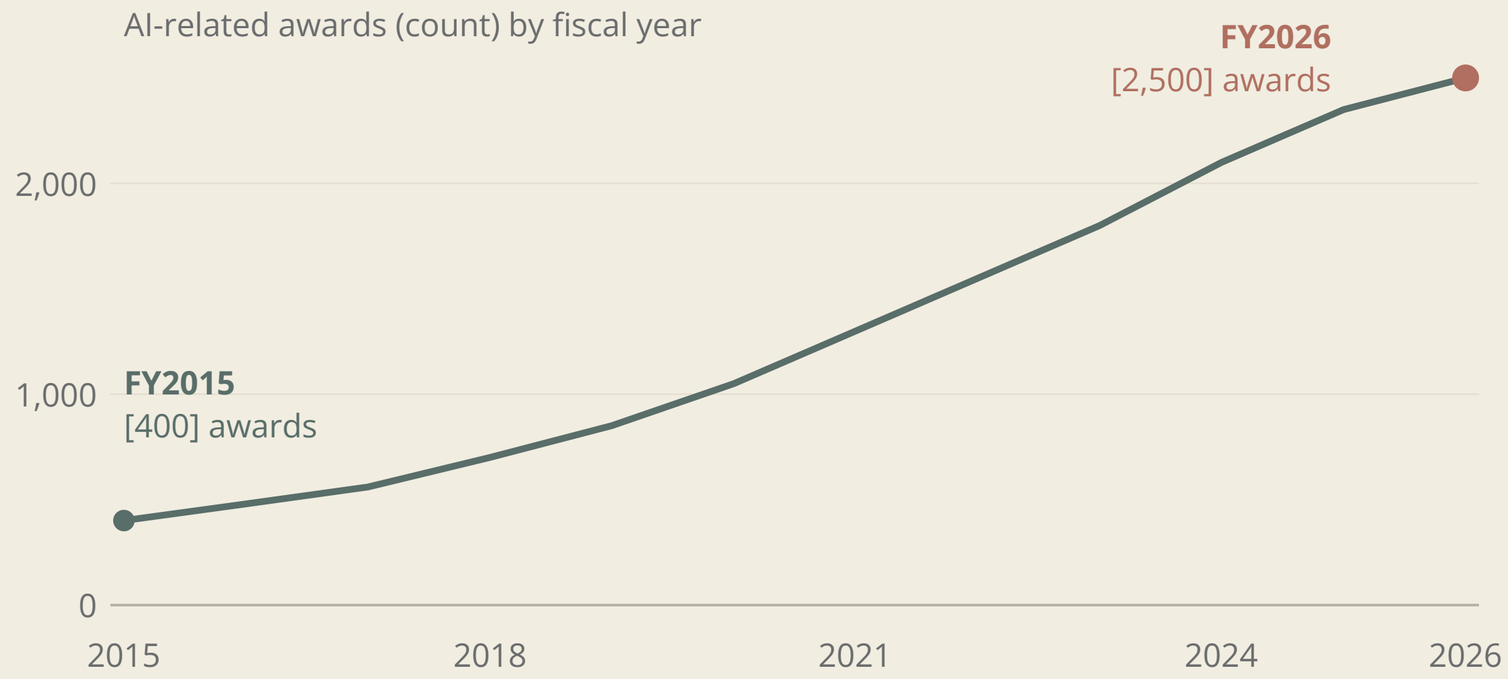
Check the labels

Hand-coded sample (Cohen's κ) and RCDC agreement

RCDC = Research, Condition, and Disease Categorization (NIH's spending categories) · **LLM** = large language model · **NLP** = natural language processing · **κ** = agreement beyond chance (1 = perfect, 0 = chance)

Then linked to iCite, OpenAlex, ClinicalTrials.gov, and PatentsView for impact

AI-related awards grew every year, about [6×] since FY2015



Steady growth, not a single spike

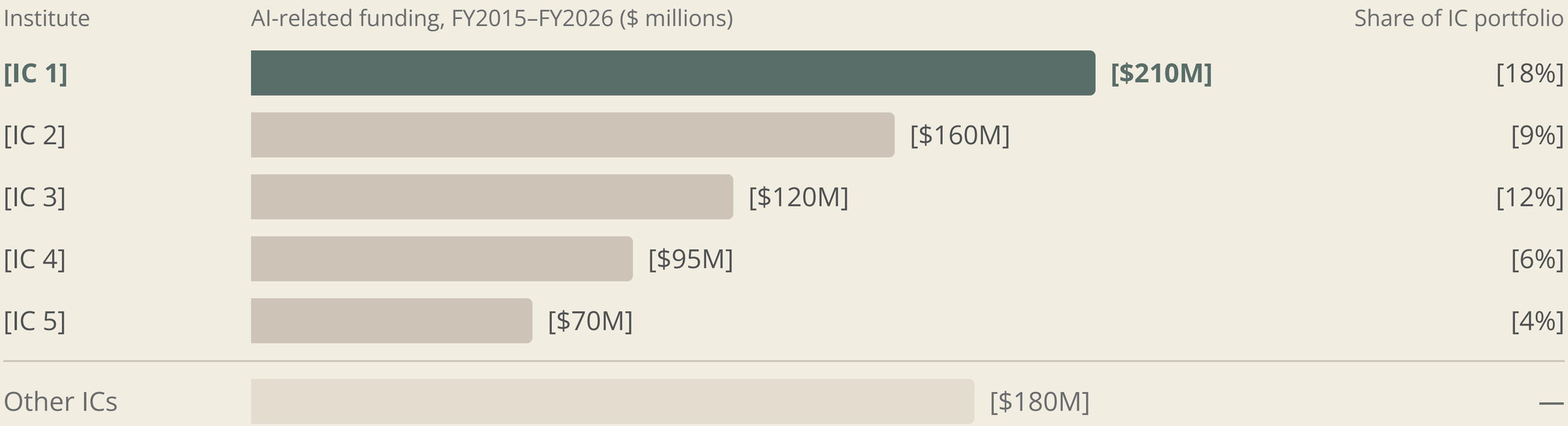
Fastest rise after FY2020 [confirm with data]

Funding follows the same shape [confirm]

Illustrative data, not results. Source when built: NIH RePORTER, taxonomy labels

WHERE AI APPEARS · BY INSTITUTE

Five institutes hold most AI funding, and [IC 1] leads



Illustrative data, not results. IC = NIH Institute or Center. Source when built: NIH RePORTER

WHERE AI APPEARS · BY METHOD AND RESEARCH AREA

Methods follow the data: vision in imaging, language models in clinical records

Awards (count)	Imaging	Genomics	Clinical & EHR	Behavioral health	Drug discovery
Machine learning	[120]	[260]	[310]	[90]	[140]
Deep learning	[340]	[150]	[120]	[40]	[110]
NLP	[20]	[30]	[280]	[110]	[40]
LLMs	[10]	[15]	[140]	[60]	[30]
Computer vision	[380]	[20]	[60]	[30]	[15]

Illustrative data, not results. NLP = natural language processing; LLM = large language model; EHR = electronic health record

Low High · 6

AI-related grants publish [more] and are cited [more] than matched grants

Publications per award



Median RCR



RCR (Relative Citation Ratio) = a paper's citations per year ÷ the expected rate for its field; **1.0 = typical NIH-funded paper.**
Matched = same institute, activity code, fiscal year, and award size band.

Illustrative data, not results. Source when built: NIH RePORTER, iCite, OpenAlex

IMPACT · TRANSLATION

AI-related grants link to trials and patents [more] often than matched grants

Awards linked to a clinical trial

AI-related



[9%]

Matched non-AI



[7%]

Awards linked to a patent

AI-related



[5%]

Matched non-AI



[3%]

Linked = the award number appears in a ClinicalTrials.gov record or in a patent's government-interest statement (PatentsView). Same matched comparison as the previous slide.

Illustrative data, not results. Source when built: ClinicalTrials.gov, PatentsView, NIH RePORTER

SO WHAT

A shared AI taxonomy lets funders plan and report AI investment

1

See it

Track AI investment by institute, method, and area in one view

2

Find gaps

Light cells in the method × area map point to areas to examine

3

Measure impact

Compare outputs and translation against matched grants

Limits: LLM label error, citation and patent lag, incomplete publication links

Next: run on real data, publish the taxonomy, refresh quarterly